

Phase 3 Test Case Report

19 / 19 automated tests passed | Combined 39 / 39 | 16 September 2026

Phase 3 tests the document vault, multi-document recalculate, MATCH / MISMATCH / NOT_AVAILABLE comparison, coverage, risk score with reasons, finalized report, SHA-256, local GENESIS hash chain, VALID / TAMPERED integrity, and ownership name extraction from Indian sale deeds. Suite: backend/tests/test_phase3.py. pytest does not call the live Sarvam API. The live UI now shows a simple risk report (score + reasons) while these APIs still prove the engine.

1. Summary

Phase 3 passed	19
Phase 1 still passing	13
Phase 2 still passing	7
Combined	39 / 39
Last duration	about 19 seconds
Harness	pytest 9.1.1 + FastAPI TestClient
Database in pytest	SQLite in-memory
Sarvam in pytest	Mocked
Integrity	SHA-256 + GENESIS ledger in Postgres; JSON copies in storage/ledger
Verdict	Phase 3 automated tests passed.

2. Automated cases

TC-30 Extract seller and buyer from executed-by sale deed

Function: test_extracts_party_names_from_executed_by_sale_deed

Steps: Run fallback_from_source on English text 'executed ... by: Sri. Ramachandra Prasad' and 'Deed in favour of: Sri. Arjun Kumar Shetty'.

Expected: seller Sri. Ramachandra Prasad, buyer Sri. Arjun Kumar Shetty, previous deed 1823/2015.

Actual: As expected. Status: Pass

TC-31 Extract party names from Kannada sale-deed cues

Function: test_extracts_party_names_from_kannada_sale_deed

Steps: Run fallback_from_source on Kannada executed-by / in-favour-of blocks.

Expected: Seller and buyer names are found. Actual: As expected. Status: Pass

TC-32 Incomplete ownership still attaches source evidence

Function: test_ownership_incomplete_attaches_source_evidence

Steps: Analyze a sale deed with survey 45/3 and no party names.

Expected: OWNERSHIP DETECTED. Evidence includes 45/3 and seller/buyer missing notes.

Actual: As expected. Status: Pass

TC-33 Indian sale deed reads seller and buyer names

Function: test_ownership_reads_names_from_indian_sale_deed

Steps: Mock Kannada OCR + English translation of the litigation sale deed. POST analyze.

Expected: Seller Sri. Ramachandra Prasad, buyer Sri. Arjun Kumar Shetty. Ownership is not DETECTED only because names were missing.

Actual: As expected. Status: Pass

TC-34 Survey and area values normalize for comparison

Function: test_values_match_normalizes_survey_prefixes

Steps: Compare Survey No. 45/3 vs Sy. No. 45/3; 1200 sq ft vs 1,200 square feet; Ramesh vs Suresh.

Expected: Survey and area MATCH after normalize. Names do not match. Actual: As expected. Status: Pass

TC-35 Multiple documents belong to one case

Function: test_multiple_documents_belong_to_one_case

Steps: Upload SaleDeed.pdf and EC.pdf to the same residential case. GET documents.

Expected: Both IDs listed. Count = 2. Actual: As expected. Status: Pass

TC-36 Matching survey numbers are MATCH

Function: test_matching_survey_numbers_are_match

Steps: Analyze sale deed 45/3 and survey sketch Sy. No. 45/3. POST recalculate.

Expected: survey_number comparison result MATCH. Actual: As expected. Status: Pass

TC-37 Different survey numbers are MISMATCH

Function: test_different_survey_numbers_are_mismatch

Steps: Sale deed 45/3 vs survey sketch 45/4. Recalculate.

Expected: MISMATCH, severity HIGH, PROPERTY_RECORD DETECTED. Actual: As expected. Status: Pass

TC-38 Different areas are MISMATCH

Function: test_different_areas_are_mismatch

Steps: Sale deed 1200 sq ft vs khata 1000 sq ft. Recalculate.

Expected: area MISMATCH with both values present. Actual: As expected. Status: Pass

TC-39 Recalculation uses all uploaded documents

Function: test_recalculation_uses_all_uploaded_documents

Steps: Analyze sale deed and nil-encumbrance EC. Recalculate.

Expected: analyzed_document_count = 2. Litigation DETECTED. Mortgage NO_ISSUE_FOUND.

Actual: As expected. Status: Pass

TC-40 Mortgage clears when EC has nil encumbrance

Function: test_mortgage_clears_when_ec_has_nil_encumbrance

Steps: Analyze sale deed only, then add EC with nil encumbrance, recalculate.

Expected: Mortgage starts NOT_VERIFIED, then NO_ISSUE_FOUND. Actual: As expected. Status: Pass

TC-41 Missing evidence remains NOT_VERIFIED

Function: test_missing_evidence_remains_not_verified

Steps: Analyze litigation sale deed only. Recalculate.

Expected: Mortgage and Approval NOT_VERIFIED. Not NO_ISSUE_FOUND. Actual: As expected. Status: Pass

TC-42 Verification coverage is calculated

Function: test_verification_coverage_is_calculated

Steps: Recalculate after one analyzed sale deed.

Expected: coverage percent 0-100, five categories, explanation is not a safety guarantee.

Actual: As expected. Status: Pass

TC-43 Risk score is deterministic (litigation = 70)

Function: test_risk_score_is_deterministic

Steps: Recalculate twice on the same litigation sale deed.

Expected: Same score both times. Score 70 HIGH. Reason includes 'Litigation on the property'.

Actual: As expected. Status: Pass

TC-44 Final report hash and GENESIS ledger

Function: test_final_report_hash_and_ledger

Steps: Analyze, POST finalize, GET reports.

Expected: version 1. report_hash = SHA-256 of canonical JSON. previous_hash = GENESIS. record_hash set. Listed report id matches.

Actual: As expected. Status: Pass

TC-45 Second report links to previous hash

Function: test_second_report_links_to_previous_hash

Steps: Finalize twice on the same case.

Expected: version 2. second.previous_hash == first.record_hash. Actual: As expected. Status: Pass

TC-46 Untouched report verifies VALID

Function: test_untouched_report_passes_integrity_verification

Steps: Finalize. GET .../reports/{id}/verify.

Expected: status VALID. report_hash_valid, ledger_valid, chain_valid all true.

Actual: As expected. Status: Pass

TC-47 Modified report verifies TAMPERED

Function: test_modified_report_fails_integrity_verification

Steps: Finalize. Change stored survey_number to TAMPERED. GET verify.

Expected: status TAMPERED. report_hash_valid false. Actual: As expected. Status: Pass

TC-48 Unauthorized user cannot access report

Function: test_unauthorized_user_cannot_access_report

Steps: Owner finalizes. Other user GET report. Other user POST recalculate. Owner recalculates missing case. Anonymous recalculate.

Expected: 403, 403, 404, 401. Actual: As expected. Status: Pass

3. Live / manual notes

TC-49 Document vault: upload, analyze, delete, and Get risk report. Pass. TC-50 Simple risk report UI shows score 70 and Because: Litigation on the property for the Kannada litigation sale deed. Pass. TC-51 Report integrity VALID with SHA-256 and ledger record on screen. Pass. TC-52 CHAIN.json under storage/ledger shows GENESIS then linked record hashes. Pass. TC-53 Agricultural / Commercial / Industrial still show Get Subscription. Pass. Missing documents still stay NOT_VERIFIED and do not add risk points.

4. Re-run

cd backend

pytest tests/test_phase3.py -v

pytest tests/test_phase1.py tests/test_phase2.py tests/test_phase3.py -v

5. Out of scope

Public blockchain (Bitcoin, Ethereum, wallets, mining), real UPI/payments, and real Google/X/phone OAuth are not Phase 3 tests. Integrity is a local SHA-256 GENESIS hash chain in PostgreSQL, also written to storage/ledger/CHAIN.json. Live Sarvam is not used inside pytest.